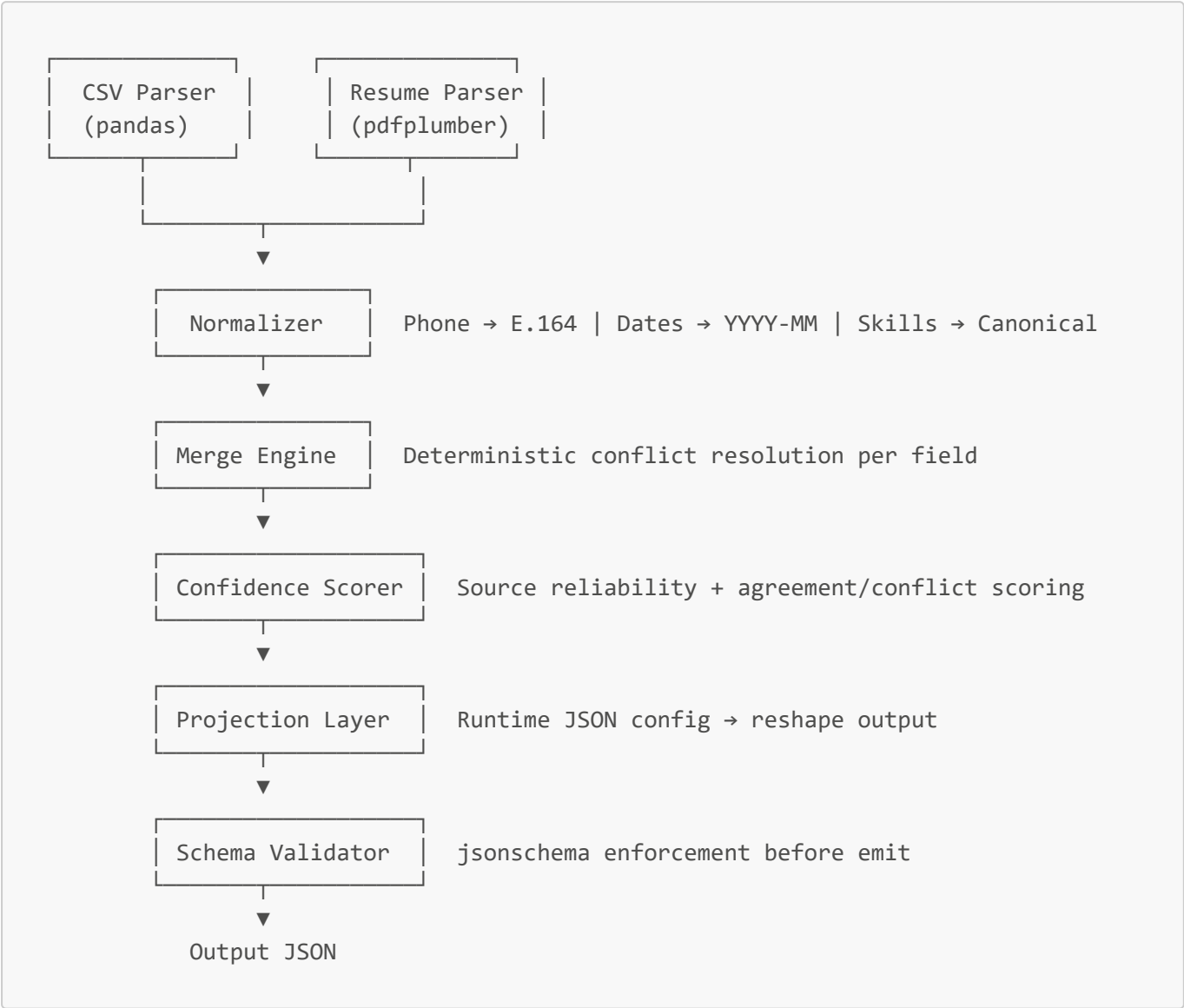


Technical Design Document: Multi-Source Candidate Data Transformer

1. Architecture & Pipeline



2. Canonical Schema & Normalization

The internal `CandidateProfile` enforces strict types. Every field is wrapped in a `TrackedField` storing: value, provenance list, confidence score.

Field	Normalization	Format
Phones	<code>phonenumbers</code> library	E.164 (+919876543210)
Dates	<code>dateparser</code> library	YYYY-MM (2022-01)
Country	ISO-3166 alpha-2	IN, US
Skills	Explicit mapping + <code>rapidfuzz</code>	Canonical (ml → Machine Learning)

Field	Normalization	Format
Links	URL prefix normalization	<code>https://linkedin.com/in/...</code>
Candidate ID	<code>uuid5(email)</code> or <code>uuid4()</code>	Deterministic UUID

3. Merge Policy & Confidence

Merge Rules: Deterministic, reproducible conflict resolution.

- **Names:** Prefer longest non-empty string ("`John`" vs "`John Doe`" → "`John Doe`").
- **Emails/Phones:** Unique aggregation (additive, never destructive).
- **Skills:** Deduplicate by canonical name, sort alphabetically for determinism.
- **Experience:** Deduplicate by (`company`, `title`) composite key.
- **Education:** Deduplicate by (`institution`, `degree`) composite key.

Confidence Scoring:

- CSV base: 0.90 | Resume base: 0.80
- Multi-source agreement: +0.05 (→ 0.95)
- Conflict detected: -0.20 (→ 0.75)
- Overall: weighted average of all non-null field scores

4. Runtime Configuration (Projection)

Clean separation between internal `CandidateProfile` and output. A JSON config controls:

- `path`: Output key name. `from`: JSON-path source (e.g., `emails[0]`, `skills[].name`).
- `on_missing`: "`null`" | "`omit`" | "`error`". Toggle `include_confidence` / `include_provenance`.
- The projector parses array indexing and subfield extraction, then validates against the target schema.

5. Edge Cases & Scope

Handled:

1. **Missing source:** A missing PDF doesn't crash the pipeline — extraction logs a warning, processing continues with available data.
2. **Empty arrays:** Requesting `emails[0]` on an empty list returns `null` (or `omit/error` per config), never an `IndexError`.
3. **Conflicting names:** Deterministically resolved (longest wins), confidence penalized to signal uncertainty.
4. **"Present" in dates:** Stored as `null` — a deliberate choice meaning "current role", not a missing value.
5. **Skill variations:** `ml`, `machine-learning`, `Machine Learning` all collapse to a single canonical "`Machine Learning`".

Descoped:

1. **Advanced NLP:** No spaCy/LLM for resume NER — heuristic section parsing is sufficient for this scope.
2. **Cross-candidate dedup:** This pipeline merges data for one known candidate. Entity resolution across candidates is a separate system.